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THE PROPORTION BY WHICH **GANYMEDE'S** MASS IS GREATER THAN THAT OF **EARTH'S MOON**

Mathematician Joseph Louis Lagrange calculated the motions of the four moons of Jupiter known to 18th-century science: Io, Europa, Ganymede, and Callisto.

IN 1765, SECLUDED IN HIS WORKSHOP IN THE UNIVERSITY OF GLASGOW, young Scottish engineer **James Watt** (1736–1819) was working on improving Thomas **Newcomen's steam engine**, which was still only in

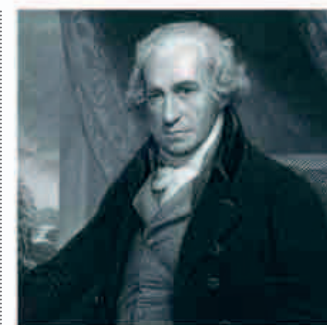
small-scale use. In Newcomen's engine, steam was let into the cylinder to push the piston up, then cold water was sprayed in and the steam condensed, creating a vacuum that pulled the piston down. The cold water

wasted a great deal of heat, so Watt added a **separate component to condense steam outside the cylinder** and avoid continuous chilling. When Watt teamed up in the 1770s with Birmingham industrialist **Matthew Boulton** (1728–1809) to put his improvements into practice, Watt's invention transformed the steam engine from a pump of limited use to the source of power that drove the Industrial Revolution.

Boulton established the Lunar Society of Birmingham, a small group of pioneering thinkers that included Erasmus Darwin, Josiah Wedgwood, and **Joseph Priestley** (1733–1804). Later Watt, too, joined them. Priestley and chemists Joseph Black and **Henry Cavendish** (1731–1810) became known as the "pneumatic chemists" for their work on air and gases. In 1765, Cavendish discovered that **hydrogen**, which he called "inflammable air," is an element, and can be made by dissolving metals in acid.

In 1766, Swiss mathematician **Leonhard Euler** accepted a new post at the Academy of Sciences in St. Petersburg, where he had spent much of his earlier career. During this year he developed key **equations for the movement of rigid bodies**—that is, any objects that keep their shape, such as a planet. Italian-born French mathematician **Joseph**

Industrial power
James Watt's revolutionary engine, incorporating a double-acting cylinder that allowed steam to escape to a separate condenser, was a key development for industry.

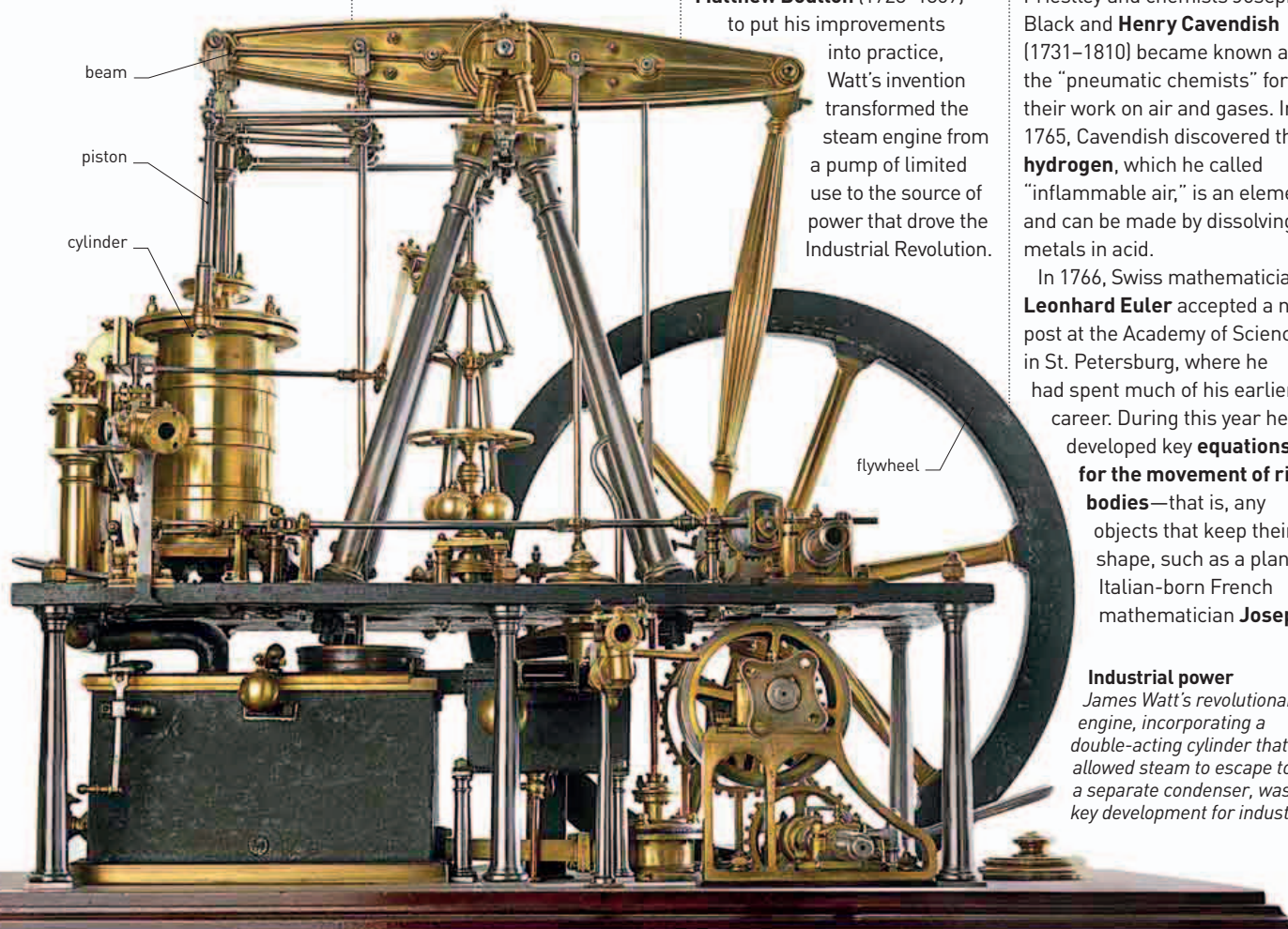


JAMES WATT
(1736–1819)

Born on January 19, 1736 in Greenock, Scotland, engineer James Watt was one of the greatest-ever inventors. His improvements to the steam engine transformed it from erratic mine pump to factory powerhouse. His many other inventions included a machine for duplicating sculptures and the world's first copying machine.

Louis Lagrange (1736–1813) took over Euler's position at the Prussian Academy of Sciences in Berlin, writing a paper on the **motion of Jupiter's moons**, only four of which were then known. It was a century later that they were given the names Io, Europa, Ganymede, and Callisto.

In Kurume, Japan, **Arima Yoriyuki** (1714–83) **calculated pi** to 29 decimal places.



1765 British engineer James Watt develops his **steam engine** with external condenser

1765 Swiss mathematician Leonhard Euler develops his theory on the **motions of rigid bodies**

1766 French mathematician Joseph Louis Lagrange analyzes the **movements of Jupiter's moons**

1766 Japanese mathematician Arima Yoriyuki **calculates pi** to 29 decimal places

1766 English chemist Henry Cavendish discovers **hydrogen**